

Review

Two-Dimensional Materials for Thermal Management Applications

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With the advances of the electronics industry, the continuing trend of miniaturization and integration imposes challenges of efficient heat removal in nanoelectronic devices. Two-dimensional (2D) materials, especially graphene and hexagonal boron nitride (h-BN), are widely accepted as ideal candidates for thermal management materials due to their high intrinsic thermal conductivity and good mechanical flexibility. In this review, we introduce phonon dynamics of solid materials and thermal measurement methods at nanoscale, and highlight the unique thermal properties of 2D materials in relation to sample thickness, domain size, and interfaces. In addition, we discuss recent achievements of thermal management applications in which 2D materials act as heat spreader and thermal interface materials based on their controlled growth and self-assembly. Finally, critical consideration on the challenges and opportunities in thermal management applications of 2D materials is presented.

Introduction

Over the past century, modern electronics has undergone significant miniaturization and high-power densification, and the trend is still continuing.^{1,2} Moore's law, which predicts that the number of transistors per square area will double in roughly every 18 months, continues to lead the pace of the semiconductor industry since it was put forward in the 1970s.³ Significantly, this prediction on the development of electronics shows a direct relationship with heat dissipation. In industrial use, thermal design power (TDP) is defined as the maximum amount of heat that a chip can generate in its working condition, which serves as an important factor for chip design and thermal management in industry. It is easily recognized from some representative Intel microprocessors as a function of time (Figure 1) that the trend of increasing density of transistors and TDP is still continuing, leading to an unavoidable increase in temperature and significant reduction of device reliability.⁴

With rapid increase in power density in modern electronics, efficient heat removal has become an emerging demand for electronic devices including communication, information, and energy storage technologies because the higher a microprocessor's cooling rate, the faster it can operate and therefore operate with better performance. Ultra-fast high-frequency devices produce a large amount of heat in small localized areas, and thus generate hotspots where the working power can be over ten times higher than its average value.⁵ As a result, deteriorated performance or even failure may occur on devices with low thermal conductivity, despite the relatively low operating temperatures of the majority of other regions.⁶

Thermal management is the effort to control the working temperature of devices such as integrated circuits (IC) by means of temperature monitoring and device cooling. As a broad definition, a thermal management setup is composed of a temperature controller, liquid/air cooling system, heat sink, and thermal interface materials

Context & Scale

Two-dimensional (2D) materials, especially graphene and hexagonal boron nitride (h-BN), have attracted great interest in thermal management as they exhibit superior thermal properties, in addition to other unique advantages of being atomically thin and mechanically flexible. Therefore, 2D materials are poised to potentially solve the increasingly severe heat dissipation problems in electronic and optoelectronic devices. Developing 2D material-based thermal interface materials with desired thermal properties has been extensively investigated. In this review, we focus on recent progress of integration of 2D materials into three-dimensional (3D) structures and their use as thermal interface materials as well as heat spreaders.

We first present the mechanism of thermal transport in 2D materials and advanced thermal characterization methods. The utilization of 2D materials with various composite forms for building phonon conduits is then reviewed. It is noteworthy that 3D structures constructed by purely 2D materials possess the highest thermal conductivity among all 3D composites. Finally, we summarize the challenges and opportunities, including interfacial engineering, 3D

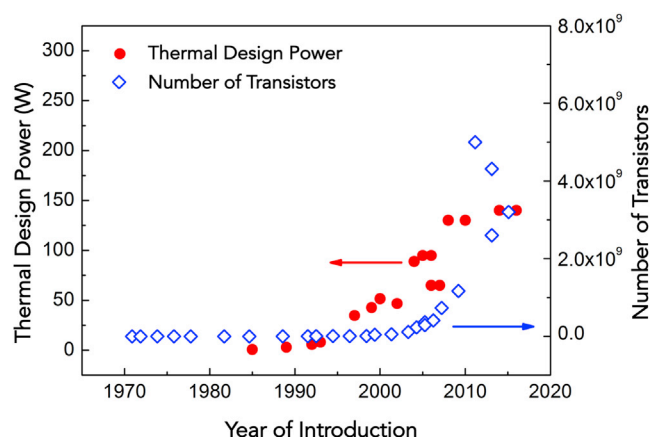


Figure 1. Timeline of Thermal Design Power and Number of Transistors for Intel Microprocessors

The data points show an overall exponential trend as a function of time consistent with Moore's law. Data were collected from Intel.⁴

(TIMs). Thermal management methodologies of hotspot removal include selection of materials with high thermal conductivity, as well as designing of heat sink and TIMs. The advance of these techniques is particularly related to the field of materials science, and is the topic of this review.

The rise of two-dimensional (2D) materials provides exciting opportunities for thermal management.^{7,8} From inception in 2004, 2D materials such as graphene, hexagonal boron nitride (h-BN), transition metal dichalcogenides (TMDCs), and black phosphorous (BP) have attracted extensive interest from researchers and engineers because of their high surface area, good mechanical strength, and tunable physical properties.^{9–12} As a result, 2D materials have shown great potential in a wide range of applications including thermal management, field-effect transistors, photodetectors, photovoltaic modules, energy storage devices, and catalysis.^{13–19} In addition, these atomically thin materials show good mechanical properties and flatness due to their 2D nature, making them promising candidates for next-generation flexible electronic devices.^{20,21} Impressively, it has been demonstrated that graphene shows ultrahigh thermal conductivity of $4,800\text{--}5,300\text{ W m}^{-1}\text{ K}^{-1}$,^{22,23} exceeding its bulk counterpart graphite, which already has a high thermal conductivity of $\sim 2,000\text{ W m}^{-1}\text{ K}^{-1}$.²⁴ These results suggest the potential of graphene as a better choice for heat spreaders than commonly used metals.^{25,26} Different from graphene, which has high thermal conductivity as well as electrical conductivity, h-BN is electrically insulating yet highly thermally conductive. These features make h-BN a superb candidate as a thermal management material for electronic devices on many practical occasions where electrical insulation is required.²⁷ From the physics point of view, 2D materials exhibit distinctive thermal properties from their bulk counterparts, such as ballistic^{28,29} and hydrodynamic³⁰ transport of phonons at $\sim 100\text{ K}$. In addition, the intriguing phonon behavior provides an opportunity to update knowledge of heat transfer at the nanoscale. In summary, 2D materials with novel phonon properties, high thermal conductivity, and good mechanical flexibility are promising for both fundamental studies of heat transfer and thermal management applications in next-generation electronics.^{31–33}

This review aims to provide an overview of fundamental phonon transport and thermal management applications of 2D materials. Basic phonon dynamics are presented first, followed by thermal measurement methods and thermal transport of 2D

structure construction, and 2D device fabrication, for future research and development of thermal management in electronics.

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materials. We then introduce the use of 2D materials as thermal management materials, especially when used as TIMs and heat spreaders. In the last section, we discuss challenges and opportunities in thermal management from a material perspective. This review could serve as an introduction for interested readers to learn the thermal properties of 2D materials and their applications in thermal management.

Heat Transfer and Thermal Measurements in 2D Materials

Phonons and Thermal Transport in Solids at the Nanoscale

The capability of a material to conduct heat can be described by thermal conductivity κ through a macroscopic expression, i.e., Fourier's law,

$$\vec{q} = -\kappa \nabla T, \quad (\text{Equation 1})$$

where q is the local heat flux and represents the amount of energy that flows through a unit area per unit time, and ∇T is the local temperature gradient.³⁴ It is widely accepted that in non-metals, heat is mainly carried by phonons, which are defined as the quanta of crystal lattice vibration.³⁵ Treating the vibrating lattice of a solid as a box of phonon gas, a simple kinetic model leads to the phonon-carried thermal conductivity,

$$\kappa = \frac{1}{3} C_v v l, \quad (\text{Equation 2})$$

where C_v is the specific heat of the lattice, v is the phonon group velocity, and l is the mean free path of phonons in the material. Related to this equation is the concept of phonon dispersion, which shows the relationship between lattice vibration frequency ω and wavevector k , $\omega(k)$, and phonon group velocity v is given by

$$v = \frac{\partial \omega_k}{\partial k}. \quad (\text{Equation 3})$$

The phonon dispersion relationship is vital for thermal transport studies as it bridges statistical atomic behavior to macroscopic heat flux. However, phonons can be scattered through anharmonicity of the lattice, including normal process (N) and Umklapp process (U) in crystal lattices,³⁶ and the latter one causes resistance to thermal transport. Notably, interfacial thermal property is particularly important for nanoscale thermal transport because nanoscale materials have high surface-to-volume ratios and are consequently more sensitive to interface than bulk materials. This makes interfacial thermal conductance, G_k , critical for thermal management in nanoscale materials. Thermal resistance at the interface is unavoidable even with atomically sharp and clean interfaces. Accordingly, the concept of Kapitza resistance is introduced as R_k , which measures thermal resistance caused by lattice mismatch across the interface.³⁷ Computational studies for solutions of Fourier's law for 2D materials were first carried out based on the BTE.³⁵ It is especially useful for large system simulations, despite the fact that the phonon dispersion of a specific material is still needed and thus limits its applications. At the same time, molecular dynamics (MD) is introduced to study the atomistic picture of thermal transport.³⁸ As MD is compatible with anharmonic interaction simulations, it is widely used to investigate fundamental thermal properties of nanomaterials.

Methods of Thermal Conductivity Measurements

Thermal measurements at the nanoscale are extremely important yet challenging. The challenges include handling and controlling small specimens, poor accuracy of temperature and heat measurements at small scales, and the introduction of complicated devices that need precise control in their manufacturing process. It makes sense to make an analogy between thermal and electric measurements, except that electrically insulating material could be easily obtained and used as a

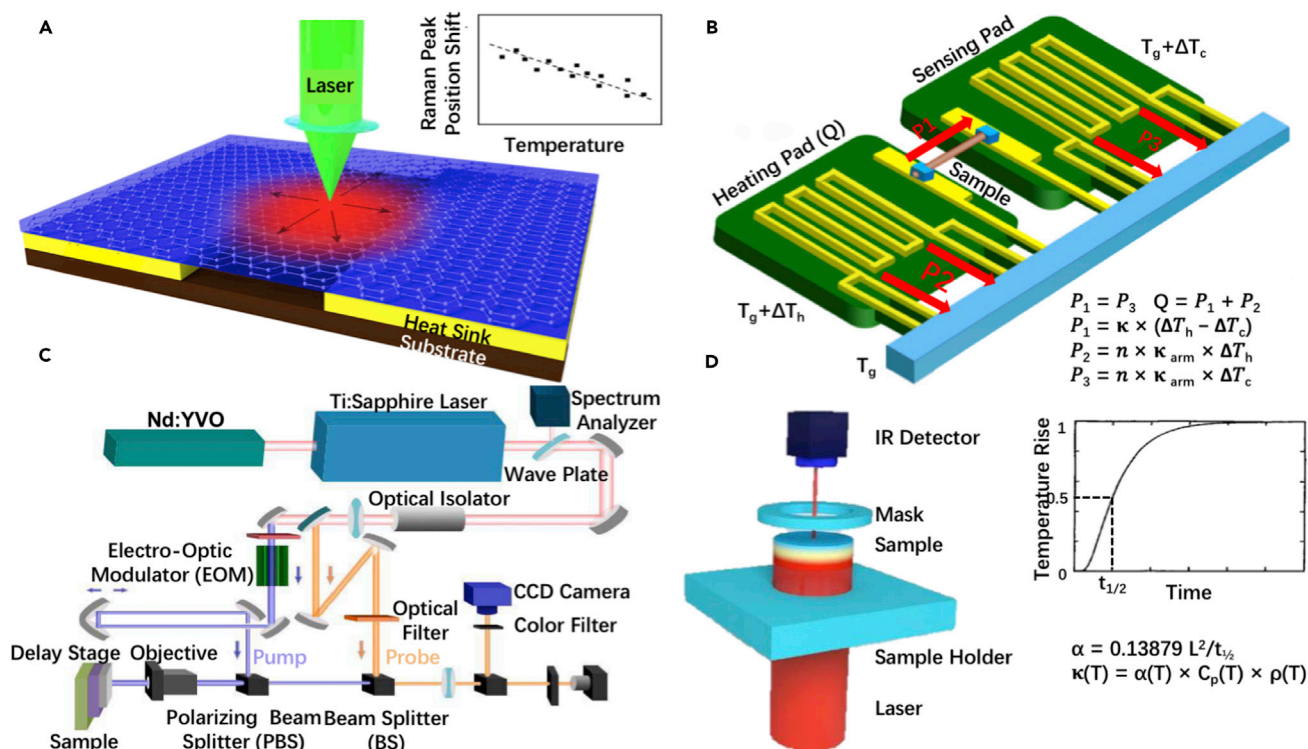


Figure 2. Scheme of Setups for Different Thermal Measurements

(A) Optothermal Raman method.

(B) Suspended-pad method.

(C) Time-domain thermoreflectance (TDTR) method, where blue and orange lines represent pump and probe laser beams, respectively.

(D) Laser flash method, where the left and right parts illustrate measurements of in-plane and cross-plane thermal conductivities.

Methods in (A) and (B) are steady-state methods, while those in (C) and (D) are transient methods.

substrate for electrical measurements, while no solid is sufficiently thermally insulating. As a result, distribution of heat flow between the sample and substrate must be carefully considered during the measurements. Instead of direct measurements of temperature, electrical^{39,40} and optical^{22,41} signals are usually measured and thermal properties are deduced from them. For more accurate measurements, heat radiation and convection in ambient air also need to be taken into consideration. To meet these challenges, various methods (Figure 2) have been developed for nanoscale thermal measurements in recent years. It should also be noted that detailed analysis of thermal measurement can be found in previous review papers.^{7,8,42}

Optothermal Raman Method. The optothermal Raman technique was first used in 2008 by Balandin et al. to measure the in-plane thermal conductivity of a suspended single-layer graphene,²² revealing a thermal conductivity as high as 4,800–5,300 W m^{−1} K^{−1}. In a typical measurement, a laser is used as a heat source and focused on the middle of a sample, causing a radial temperature gradient in the basal plane of 2D materials (Figure 2A). Because the peak shifts in Raman spectrum show linear relationship with the temperature change,⁴³ one can connect the temperature rise of the sample surface with the laser heating power, and the sample thermal conductivity can be calculated. It should be noted that the laser power absorbance of graphene with wavelength dependence is determined in different ways. In Balandin's work, a 488-nm laser was chosen for a clear Raman signal as

well as efficient heating effect, and the absorbed power for a specific laser was derived through an experimentally measured integrated Raman intensity in comparison with bulk graphite.²² Some results were based on the difference between transmitted laser power through empty holes and graphene flakes,⁴⁴ or an assumption of 2.3% absorbance of laser power by graphene;⁴⁵ therefore, the observed results with data scattering are not surprising.

In addition to suspended graphene, thermal conductivity of substrate supported graphene was later measured by an optothermal Raman method.^{19,25,44,45} As this method is optically based and non-destructive to the sample, it has been expanded to measure the thermal conductivity of other 2D materials including h-BN,⁴⁶ MoS₂,^{47,48} and WS₂.⁴⁹ However, it should be noted that there are relatively high experimental errors in the optothermal Raman method. Such errors could result from inaccurate estimation of optical absorption of sample, uncertainty of laser spot size calibration, or the unavoidable heat dissipation to the surrounding environment and interfacial thermal conductance of supported area of samples. A modified method has been developed by carrying out measurements in a vacuum, which eliminates the influence of heat loss of surrounding gas.⁵⁰ Considering that the coefficient between Raman peak shift and temperature change is usually at the level of $\sim 0.01 \text{ cm}^{-1}/\text{K}$ for graphene, it is very challenging to determine both Raman shift and temperature change with high resolution.⁴³ In addition, detailed calculations are needed to further investigate physical processes behind the relationship of Raman peak shift and the sample temperature.

Suspended-Pad Method. A micropad method was developed to measure thermal conductivity of carbon nanotubes (CNTs) as early as 2001.³⁹ The device (Figure 2B) is made of two silicon nitride (SiN_x) membranes, with a platinum (Pt) serpentine line patterned on each pad and used as microheaters and thermometers, providing Joule heating and calibrating temperature of the two pads, respectively. The pads are suspended by six SiN_x arms coated with Pt for electrical and thermal conductivity measurements. The sample, which is usually a nanoribbon or nanowire, is then transferred by a sharp tip under a microscope to bridge the two micropads. The sample is then bonded onto a Pt electrode by focused ion beam to ensure Ohmic contacts and minimize thermal resistance. Heat transfer in the sample can be calculated from Joule heating of the heated pad and temperature rise of the other, sensing pad.

This suspended-pad method has also been used to measure the thermal conductivity of 2D materials including graphene, h-BN, MoS₂, BP, and Bi₂Te₃.^{51–56} This method shows straightforward and reliable calibration on temperature with high resolution ($\sim 0.05 \text{ K}$),⁵⁷ which makes it a powerful technique for in-plane thermal conductivity measurements. However, the accuracy of this technique can be significantly affected by polymer residues on 2D materials,^{58,59} and the tip-assisted sample transfer would be technically challenging and time consuming. The device fabrication process is also complicated.⁵⁵

Time-Domain Thermoreflectance Method. The time-domain thermoreflectance (TDTR) method is a “pump-probe” optical technique to measure thermal transport at nanoscale thickness. Although the first report can be dated back to 1986,⁴¹ it was extensively studied only in recent years with the rise of nanomaterials.^{60,61} In a typical measurement, a mode-locked laser is split into a “pump” beam and a “probe” beam. The pump beam produces a temperature rise of $\sim 3 \text{ K}$ near the metallic surface (either the sample is metallic or is coated with a thin metal film). The probe beam, which is in the middle of pump, can monitor surface temperature

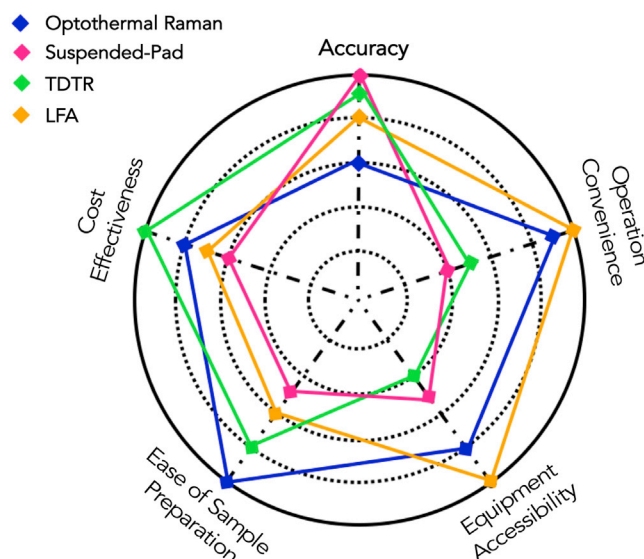


Figure 3. Comparison of Four Commonly Used Methods for Thermal Conductivity Measurements, Graded in Terms of Their Relative Accuracy, Operation Convenience, Equipment Accessibility, Ease of Sample Preparation, and Cost Effectiveness

with the change of optical reflectivity. The phase and amplitude of the probe are measured as a function of delay time between the pump and probe, and output by a lock-in amplifier (Figure 2C). A mathematical fitting is then conducted to extract thermal parameters. The resolution of heat penetration at the nanoscale makes TDTR a powerful method for interface thermal transport measurement,⁶² among which the most intensively studied sample is graphene/metal interface.^{63–68} TDTR has also been reported to be able to measure cross-plane thermal conductivity of 2D materials such as BP⁶⁹ and WSe₂.⁷⁰

Other Methods. Besides the methods discussed above, there are some other commercial instruments based on micrometer-scale resolution that could be modified for thermal measurements at the nanoscale. For example, a scanning thermal microscope (SThM) is a specially designed scanning probe microscope that gives a temperature mapping result based on a sharp scanning tip. It was first used for thermal conductivity measurement of CNTs in 2000.⁷¹ Null point scanning thermal microscopy (Np SThM) was modified by measuring the local heat transported from the sample to the tip only, minimizing the impact of heat transfer in the air gap between the sample and tip.⁷² With this method, a residue-free suspended graphene was measured and showed a thermal conductivity of $\sim 2,400 \text{ W m}^{-1} \text{ K}^{-1}$ at room temperature.⁷³ For bulk materials, one of the most widely used commercial devices is the laser flash apparatus. In this setup, the sample is cut to a specific size and heated by a pulse laser, and the thermal diffusivity can then be measured by manipulation of the thickness of the sample and thermal diffusion time (Figure 2D). It is a convenient way for thermal measurements provided that the sample size meets the requirements,^{28,74} and is especially useful for characterization of thermal management materials for real application. The advantages and disadvantages of different thermal conductivity measurement methods are summarized in Figure 3.

Thermal Transport in 2D Materials

Phonon Dispersion. In 2D materials, because most of them are semiconductors, heat is mainly transported by phonons due to the absence of high density of free

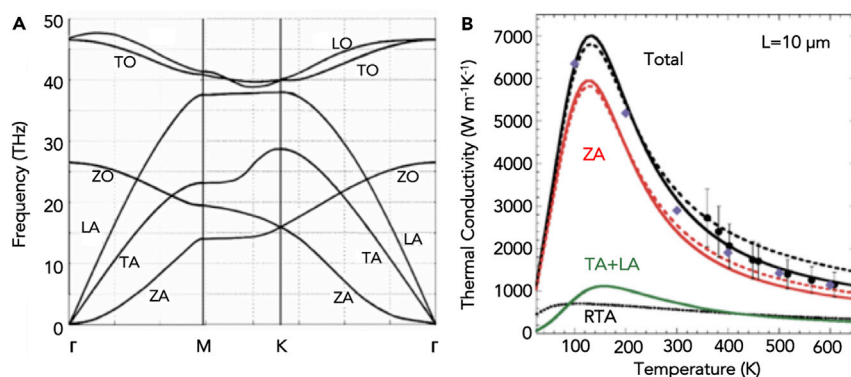


Figure 4. Theoretical Investigation of Phonons in Graphene

(A) Phonon dispersion of graphene by the valence-force field method. TA, transverse acoustic phonons; LA, longitudinal acoustic phonons; ZA, out-of-plane acoustic phonon branch; LO, longitudinal optical phonons; TO, transverse optical phonons; ZO, out-of-plane optical phonons. Reprinted from Nika et al.,³⁵ with permission. Copyright 2009, American Physical Society.

(B) Calculated thermal conductivity and the contributions of different phonon modes for graphene with a channel length of $L = 10 \mu\text{m}$. The solid black curve⁸⁵ and dashed black curve⁸³ give theoretical results while black circles are experimental data for single-layer suspended graphene.⁵⁰ The lower dotted black curve gives thermal conductivity by relaxation time approximation (RTA). Reprinted from Lindsay et al.,⁸⁵ with permission, Copyright 2014, American Physical Society.

electrons. Even in graphene, which is a semi-metal, phonons rather than electrons are the main heat carriers. Due to the strong covalent sp^2 bonding between carbon atoms, heat conduction is very efficient in its basal plane.⁷⁵ The much weaker van der Waals forces between the atomic layers compared with the covalent bonding in the basal plane leads to very anisotropic thermal conductivity for 2D materials between the x-y plane (in-plane) and along the z direction (cross-plane).⁷⁶ It was also theoretically predicted that some 2D materials could show anisotropic thermal properties in the basal x-y plane,^{77,78} which was later experimentally observed in BP.^{55,79} Klemens has calculated the intrinsic in-plane thermal conductivity of graphite^{75,80} and found it to be $\sim 1,910 \text{ W m}^{-1} \text{ K}^{-1}$ at room temperature, which is close to the previous experimental result of highly oriented pyrolytic graphite of $1,850 \text{ W m}^{-1} \text{ K}^{-1}$.⁸¹ In this model, out-of-plane acoustic phonon branch (ZA) of graphite was neglected due to its low phonon velocity and high Gruneisen factor γ ,⁸² while assuming that heat is dominantly carried by transverse acoustic phonons and longitudinal acoustic phonons (Figure 4A). However, this assumption was challenged by later studies, arguing that the ZA phonons are dominant in thermal conduction in 2D materials including graphene and h-BN.^{83,84} This behavior is attributed to a selection rule that restricts the phonon scattering of ZA phonons,⁸³ and shows consistency with experimentally measured results of SiO_2 supported graphene (Figure 4B).^{50,52,85} Overall, this has been updated by recent papers reporting that ZA phonons dominate in heat transfer in 2D materials in the in-plane direction. However, experimental investigation of the roles of different phonons in thermal conduction is still needed for a full understanding of thermal conduction physics of 2D materials.

Effect of Thickness. Thickness-dependent thermal conductivity of suspended graphene was first reported by Ghosh et al.⁸⁶ using the optothermal Raman method, and the results show a drop from $\sim 2,800$ to $\sim 1,300 \text{ W m}^{-1} \text{ K}^{-1}$ as the number of graphene layers increases from 2 to 4. However, when graphene is supported by amorphous SiO_2 , it shows rising thermal conductivity in parallel with increasing thickness.⁸⁷ The difference is attributed to distinct limitations of phonon transport in each case. In the suspended case, thermal transport is limited by intrinsic

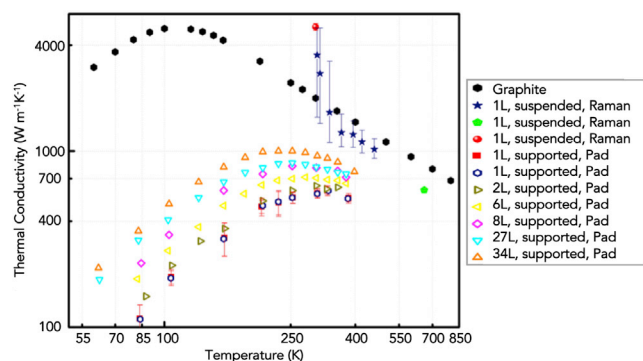


Figure 5. Experimental Results of In-Plane Thermal Conductivity of Graphene as a Function of Temperature and Different Number of Layers

The optothermal Raman technique by Balandin et al.,²² Lee et al.,⁸⁸ and Faugeras et al.,⁴⁵ and the suspended-pad method by Seol et al.⁵² and Sadeghi et al.⁸⁷ are included. Bulk graphite by Touloukian²⁴ is shown for comparison. Only results from mechanically exfoliated graphene are presented here, because other methods may produce graphene with lower quality than that of mechanically exfoliated ones.

properties, and an increased Umklapp scattering was discovered in thicker-layer graphene due to crystal anharmonicity; in contrast, in the supported case the extrinsic property is dominant, as the phonon-interface scattering leads to a suppression of phonon modes, especially the ZA modes, which would contribute most to the thermal transport.^{83,88} The above measurements are summarized in Figure 5, and more details of phonon behaviors deciding thickness-dependent thermal conductivity of graphene can be found elsewhere.⁷ Meanwhile, the reduction of thermal conductivity at small thicknesses has been found in suspended h-BN⁵¹ and MoS₂,⁵⁴ and was explained by increasing phonon scattering with polymer residues on the surface.⁵¹ Further investigation of thickness-related thermal conductivity measurements is needed to elucidate these effects, as it could provide fundamental knowledge on thermal properties of 2D materials and their applications in thermal management.

Effect of Domain Size. Intensive experimental and theoretical efforts have also been directed toward the effect of domain size on thermal properties. In the Klemens model, in-plane thermal conductivity of graphite shows a logarithmic divergence with the grain size.⁷⁵ Based on an ideal case where heat transfer is only inhibited by Umklapp scattering, simulation study indicates that the thermal conductivity of graphene can exceed the bulk limit provided that the flake domain size is up to a few micrometers.⁸⁹ This trend has been confirmed by a suspended-pad measurement, showing that the thermal conductivity keeps logarithmically increasing even if the sample length reaches 10 μm , much longer than the mean free path.⁵³ For larger domain sizes ($\sim 30 \mu\text{m}$), a finite value was discovered by theoretical study.⁹⁰ In real cases, phonons would be scattered by defects besides the Umklapp scattering.⁹¹ Nevertheless, the theoretical studies point to ways of improving the thermal conductivity by growing large crystals of 2D materials.⁹²

Heterostructures and Interfaces. Van der Waals heterostructures assembled with 2D materials has attracted great attention because of their interesting science and potential applications in electronics, optoelectronics, and thermal managements.⁹³ Both vertical and lateral 2D heterostructures have been synthesized, with an atomically flat surface three times smoother than SiO₂ (Figure 6A), and a sharp edge that is observed in high-angle annular dark field (HAADF) transmission electron

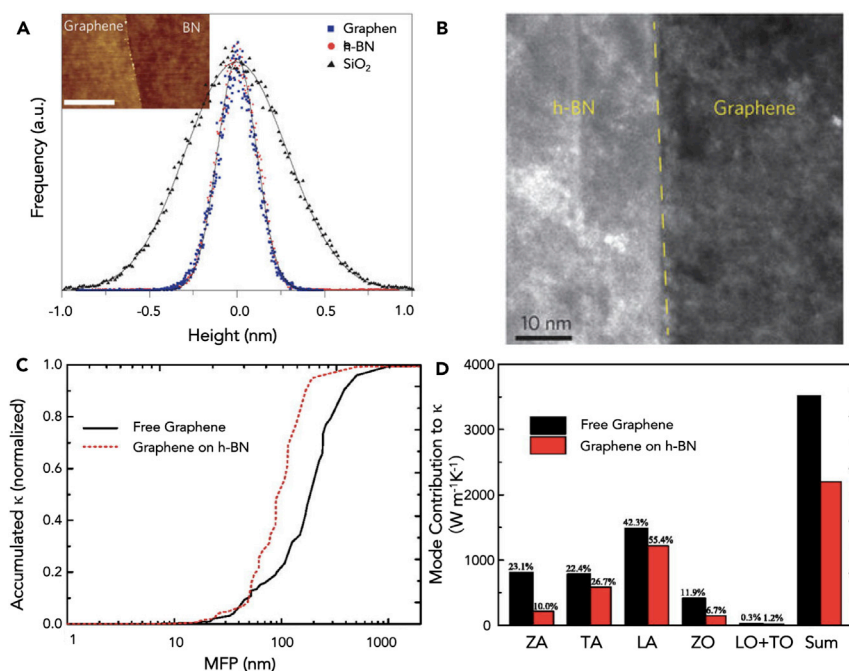


Figure 6. Characterization and Calculated Thermal Transport Property of Graphene/h-BN Heterostructures

(A) Histogram of the surface roughness of SiO₂ (black triangles), h-BN (red circles), and graphene-on-BN vertical heterostructures (blue squares) measured by atomic force microscopy (AFM). Solid lines are Gaussian fits to the distribution of height. Inset: a high-resolution AFM image showing graphene on an h-BN substrate. Reprinted from Dean et al.,⁹⁶ with permission. Copyright 2010, Nature Publishing Group.

(B) A HAADF TEM image of a graphene/h-BN lateral heterostructure. The left part is h-BN and the right part is graphene. Comparison of thermal conductivity of free and h-BN-supported graphene is shown. Reprinted from Liu et al.,⁹⁴ with permission. Copyright 2013, Nature Publishing Group.

(C) Thermal conductivity as a function of mean free path (MFP) of graphene and h-BN. Reprinted from Zou and Cao,⁹⁸ with permission. Copyright 2017, AIP Publishing LLC.

(D) Phonon mode contribution to thermal conductivity of free graphene (black) and graphene on h-BN (red). Reprinted from Zou and Cao,⁹⁸ with permission. Copyright 2017, AIP Publishing LLC.

microscopy (TEM) image as shown in Figure 6B.^{94,95} Among these, graphene/BN vertical heterostructures have been extensively studied since they show the highest charge carrier mobility of graphene supported on any substrate.⁹⁶ Meanwhile, phonon and thermal studies of heterostructures are of great interest for nanoelectronic applications.⁹⁷

Interface thermal conductance of graphene/h-BN interface was measured to be $\sim 52 \text{ MW m}^{-2} \text{ K}^{-1}$ by the optothermal Raman method, higher than that of MoS₂/h-BN interface ($17 \text{ MW m}^{-2} \text{ K}^{-1}$) due possibly to strong cross-plane transmission of phonon modes at the interface.⁹⁹ However, it was measured that the interface thermal conductance of graphene/SiO₂ interface is roughly equal to or higher than that of graphene/h-BN.^{100,101} Although the inconsistency is dominantly attributed to experimental uncertainties for different measurement methods and variation in sample quality difference, it is still intriguing to explore intrinsic thermal transport properties from the perspective of lattice configurations at the interface.¹⁰² It is suggested that intercoupling between graphene and h-BN is not strong, thus leading to failure of the symmetry-based selection rule, significantly reducing the lifetime of ZA phonons of supported graphene (Figures 6C and 6D).⁹⁸ Interfacial reinforcement

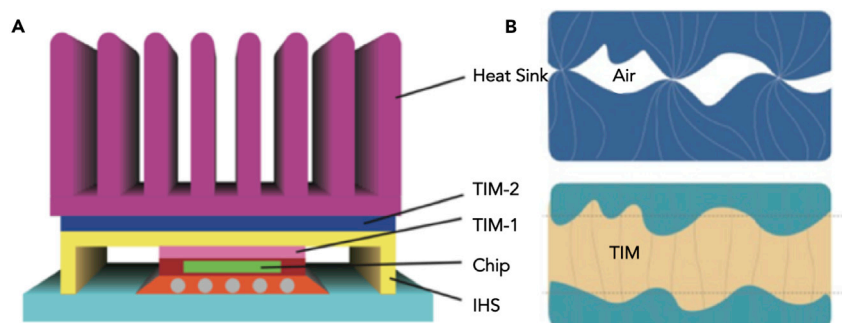


Figure 7. Schematic of an IC and Role of TIMs

(A) Illustration of a typical IC package with two TIMs and one integrated heat spreader (IHS). The TIMs are applied between chip/IHS and IHS/heat sink interfaces and defined as TIM-1 and TIM-2, respectively.

(B) Schematic of a TIM. An interface in the absence of TIM (top) shows conjunct heat flux, with heat transfer at limited solid contact areas.

can be achieved by stacking for more C-B matching to reduce interfacial space,¹⁰³ or specifically introducing 5/7 carbon defects to relax the lattice mismatch.¹⁰⁴ However, most work on this topic is based on computational methods and corresponding experimental investigations are urgently required.

Other Factors. In addition to the above factors that have significant influence on the thermal conductivity of 2D materials, other factors, including isotopic modification,¹⁰⁵ ion intercalating,¹⁰⁶ tensile strain,¹⁰⁷ defect engineering,¹⁰⁸ and functionalization,¹⁰⁹ can also be used to engineer the thermal conductivity of 2D materials. These are shown to exist mainly on graphene, but in principle they should work for other 2D materials as well.

2D Materials for Thermal Management

2D Materials as Fillers for Polymer-Based TIMs. A typical thermal management system in IC chips is depicted in Figure 7A. It contains a heat sink, an integrated heat spreader (IHS), and two TIMs named as TIM 1 and TIM 2. A TIM is used to connect two solid materials together by filling in the air gap between them, and thus reduces the interfacial thermal resistance (Figure 7B). In accordance with the package structure, TIMs can be categorized into several groups including polymer-based composites, solders, and other materials.¹¹⁰ Among them, polymer-based composites with epoxy, silicone, poly(methyl methacrylate) (PMMA) and poly(vinyl alcohol) (PVA) matrix are most widely used due to their good adaptability to versatile solid interfaces. While polymers themselves generally have low thermal conductivities ($<0.5 \text{ W m}^{-1} \text{ K}^{-1}$), fillers with relatively high thermal conductance capabilities such as ceramics,^{111–113} carbon materials,^{114,115} and metals¹¹⁶ are incorporated, exhibiting overall thermal conductivities in the range of $1\text{--}10 \text{ W m}^{-1} \text{ K}^{-1}$.¹¹⁷ 2D materials such as graphene^{118–120} and h-BN^{27,121–123} were reported to be promising candidates for highly thermal conductive fillers of polymer-based TIMs. These TIMs could also be applied on energy devices including batteries and solar cells.^{124–126}

Achieving uniform dispersion of 2D materials in a polymer matrix is of great significance in fabricating high-performance polymer-based TIMs. Typically, a TIM with high thermal conductivity requires high filler loading (up to 90 wt%) to exceed the percolation threshold, i.e., forming a thermal conductive path inside the polymer matrix.¹²⁷ High loading is also beneficial as it increases the surface area of highly thermal conductive fillers, since the practical indicator of heat removal ability, the

thermal conductance, scales linearly with both thermal conductivity and cross-section area. However, overloading of fillers could result in a loss of mechanical properties of composites; therefore, minimizing filler loading while keeping high thermal conductivity is always a preference for practical use, but remains a technical challenge. To achieve this, one has to make fillers well dispersed in the polymer matrix so that the interface thermal resistance between filler and matrix would be reduced, while the process is highly related to the preparation methods of materials. 2D material fillers are generally prepared by a liquid exfoliation process, including chemically modified exfoliation for graphene oxide (GO)^{128,129} and functionalized h-BN nanosheets,^{130,131} and direct sonication of pristine bulk materials in solvents.^{132–136} Exfoliation of chemically modified layered materials often yields good dispersion of 2D materials with high concentration, but inevitably introduces defects in the materials. On the other hand, direct exfoliation of pristine bulk materials can produce 2D materials with high quality, although the prepared samples are difficult to disperse in the polymer matrix at high concentrations because of their high cohesive energy of van der Waals force.^{137,138} Therefore, there is a trade-off between preparing high-quality fillers and building a good filler-matrix interface. In the following, methods of reaching high intrinsic thermal conductivity and enhancing the thermal interface are discussed separately.

Improvement in dispersity is very important to achieve uniform and stable dispersion of 2D fillers into a polymer matrix and consequently improve the thermal conductivity of the system. Chemical modification of 2D materials functionalizes the 2D nanosheets and therefore improves their dispersity in polymer matrices. Covalent bonding can be introduced by grafting functional groups or molecules including amines,¹³⁹ silane,¹⁴⁰ and polymer monomers¹⁴¹ onto 2D materials. However, this method is used at the expense of reducing the intrinsic thermal conductivity of 2D materials.¹⁴² Non-covalent functionalization by solvent-soluble molecules is an alternative method. For example, benzene derivatives are usually used to introduce π - π stacking with graphene while the other side of the molecule is compatible with the polymer matrix,^{120,143} leading to a system with improved interface thermal transport while not degrading the intrinsic thermal conductivity of graphene itself. Methods to modify graphene and GO have been well developed and reviewed in recent publications.^{144,145} At the same time, mechanical techniques can be introduced for improved dispersion and high loading of 2D materials in polymer matrices. An external force can be applied by means of tip sonication,¹⁴⁶ vacuum filtration mixing,^{147,148} hot-pressing,¹⁴⁹ and so forth.

Building phonon conduits is another important aspect for enhancement of the filler-polymer interface, which can be realized by the formation of hybrid structures of low-dimensional materials including graphene, h-BN, and CNTs. It was discussed in a previous section that graphenes with larger domain sizes show higher thermal conductivity than those with smaller ones. 2D materials with large domain sizes are therefore desirable, because they form a filler structure that reaches the thermal percolation threshold with less interface area.¹⁵⁰ However, solely applying fillers with large sizes may fail to form a complete heat pathway in the whole polymer matrix because of limited overlapping areas between the nanosheets of 2D materials. Therefore, the concept of building hybrid structures has been suggested, with microsize and nanosize fillers functioning as brick and cement for phonon conduits, respectively. Hybrid fillers with graphene and CNTs have attracted extensive attention since its first report in 2008.¹¹⁸ A high loading of graphene and CNTs with synergic effect was later reported with a loading of 20 vol% each, showing a thermal conductivity of $6.18 \text{ W m}^{-1} \text{ K}^{-1}$, which is higher than that of 50 vol% individual

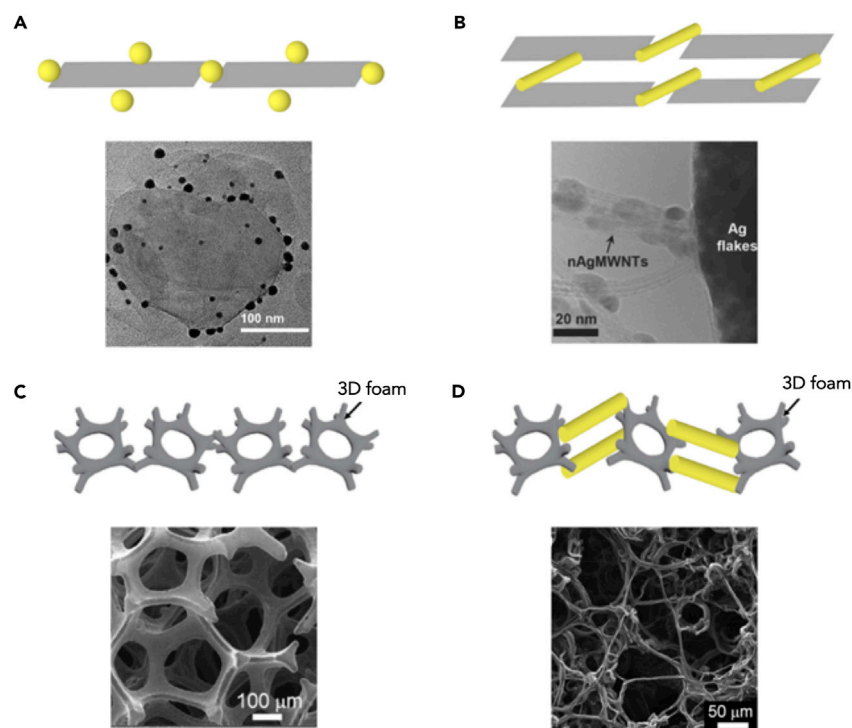


Figure 8. 2D Materials in Polymer-Based TIMs with Hybrid and Interconnected Structures

(A–D) Schematic and experimental results are shown at top and bottom of each column separately. (A) 2D nanosheets combined with nanoparticles. Reprinted from Wang et al.,¹⁵² with permission. Copyright 2016, Nature Publishing Group.

(B) 2D nanosheets combined with 1D nanotubes. Reprinted from Suh et al.,¹⁵³ with permission. Copyright 2016, Wiley.

(C) 3D foam structure. Reprinted from Kholmanov et al.,¹⁵⁵ with permission. Copyright 2015, American Chemical Society.

(D) 3D foam structure with 1D nanotube connection. Reprinted from Kholmanov et al.,¹⁵⁵ with permission. Copyright 2015, American Chemical Society.

CNTs or graphene.¹⁴⁶ Building a micro-nano-size hybrid structure with different shapes of fillers shows improved thermal conductivity, and these examples include graphene flakes with Ag particles,^{151,152} h-BN flakes with SiC nanowires,¹⁴⁸ and Ag flakes with CNTs¹⁵³ (Figures 8A and 8B). Simulations show that a microsize filler is substantial for reaching percolation threshold due to its large contact surface areas with polymer matrix,¹⁵⁴ which is consistent with experimental results mentioned above.

Another effective method to build phonon conduits is to form 3D interconnected structures. In recent years, construction of thermally conductive 3D interconnected networks has become a new research frontier in TIM design. Compared with methods mentioned above, filling a highly thermally conductive 3D structure requires lower filler loading to reach percolation threshold. Several methods have been reported to synthesize 3D macroforms of graphene and h-BN, such as direct growth by chemical vapor deposition (CVD) using Ni foam as a template,¹⁵⁶ and self-assembly of GO.^{157,158} CVD produces graphene foams (GFs) with a highly crystalline structure. A GF/wax composite shows improved thermal conductivity of $2.13 \text{ W m}^{-1} \text{ K}^{-1}$, which is up to 18-fold enhancement compared with bare wax, at an ultralow GF loading of 1.23 vol%.¹⁵⁹ This value can be further increased by 1.8-fold when the porous space of GFs is occupied by long CNTs, forming

Table 1. Comparisons of Thermal Conductivities of Polymer-Based TIMs with Different Fillers

Fillers	Loading	K ($\text{W m}^{-1} \text{K}^{-1}$)	Method	Reference
Filler Modification				
rGO	25 vol%	6.44	–	Yu et al. ¹¹⁵
rGO	20 wt%	5.8	silane functionalized	Ganguli et al. ¹⁴⁰
Graphene	10 vol%	5.1	sodium cholate functionalized	Shahil et al. ¹¹⁹
Graphene	10 wt%	1.53	pyrene derivatives functionalized	Song et al. ¹⁴³
rGO	4 wt%	1.91	pyrene derivatives functionalized	Teng et al. ¹²⁰
h-BN nanotube	30 wt%	2.77	CVD grown, silane functionalized	Huang et al. ¹²³
Forced Mixing				
h-BN	94 wt%	6.9 (κ_l)	[PVA] vacuum filtrated	Zeng et al. ¹⁴⁷
GO	10 wt%	0.244 (κ_l)	hot pressed	Ding et al. ¹⁴⁹
Hybrid Filler				
Graphene + CNT	7.5 wt% + 2.5 wt%	1.75	GNP, single-walled carbon nanotube	Yu et al. ¹¹⁸
Graphene + CNT	20 vol% + 20 vol%	6.31	GNP, multiwalled carbon nanotube	Huang et al. ¹⁴⁶
Ag flake + CNT	35.8 vol% + 2.3 vol%	160	Ag-nanoparticle-functionalized CNT	Suh et al. ¹⁵³
h-BN + SiC nanowire	95 wt%	21.7	[PVA] vacuum filtrated	Yao et al. ¹⁴⁸
3D Interconnected Structure				
Graphene	1.23 vol%	3.44	[wax] CVD grown, annealed	Ji et al. ¹⁵⁹
Graphene + CNT	15.8 vol% (in total)	4.09	[erythritol] CVD grown, annealed	Kholmanov et al. ¹⁵⁵
h-BN	1.5 vol%	1.03	[PMMA] CVD grown	Xue et al. ¹⁶⁰
rGO	0.92	2.13	aligned freeze drying, 400°C reduced	Lian et al. ¹⁶³
h-BN	9.29	2.8	aligned freeze drying	Zeng et al. ¹⁶²

Epoxy resin was used as matrix unless otherwise specified.

concentrated heat transfer channels in GF/CNT/polymer composites (Figures 8C and 8D).¹⁵⁵ Similarly, 3D h-BN foam has been grown by Ni foam-templated CVD,¹⁶⁰ and showed nearly 3-fold improvement of thermal conductivity in h-BN/PMMA composites.¹⁶¹ Based on GO dispersion, the solution method is an alternative way to produce 3D connected filler networks. Using an ice-templated method, an alternative way to produce 3D connected filler networks was accomplished by promptly contacting the hydrogel with cold medium, creating a 3D network with preferred structure orientation.^{162,163} Typical thermal conductivities of polymer-based TIM with 2D materials fillers are summarized in Table 1. Solution-induced aerogel is easy to be scaled up at relatively low cost, despite the fact that the functionalization of graphene and BN nanosheets degrades their high intrinsic thermal conductivity. The aerogel needs to be annealed properly in order to recover the thermal conductivity of pristine graphene/h-BN as much as possible, while retaining the mechanical strength of as-prepared monolith. Efforts have been devoted to improving the loading fraction by condensing the aerogel,¹⁶⁴ or controllable synthesis for anisotropic thermal conductive frameworks.¹⁶⁵ Many other factors are also essential for real applications such as clamping pressure, cycle ability, and liquid leakage. These indicators are primarily related to industrial technology and are therefore beyond the scope of this paper. However, it is worth noting when designing real thermal management applications.

Direct Use of 2D Materials and Their 3D Assembly for Thermal Management

Graphene and h-BN filled TIMs have shown significantly enhanced thermal conductivity compared with commercial ones. However, the overall performance is still far

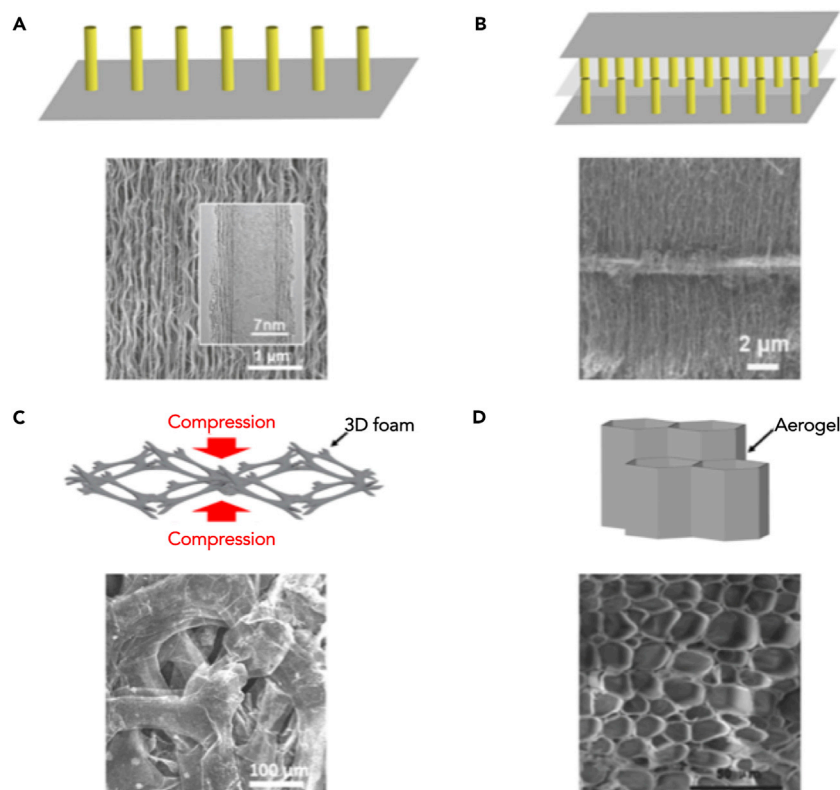


Figure 9. 2D Materials and Their 3D Structure for TIMs

(A–D) Schematic and experimental results are shown at top and bottom of each column separately.

(A) 2D nanosheet with vertically grown 1D nanotubes. Reprinted from Wang et al.,¹⁶⁷ with permission. Copyright 2008, IOP Publishing.

(B) 1D pillared/2D layered structure. Reprinted from Tang et al.,¹⁷⁴ with permission. Copyright 2014, Elsevier.

(C) Compressed 3D foam. Reprinted from Loeblein et al.,¹⁷⁵ with permission. Copyright 2017, American Chemical Society.

(D) Vertically aligned aerogel. Reprinted from Zeng et al.,¹⁶² with permission. Copyright 2015, Wiley.

from satisfactory because of the low thermal conductivity of polymer matrices. On the other hand, using 2D materials as fillers fails to exploit their flexibility. Therefore, directly using 2D materials as thermal management materials has become a pursuit.

3D-Interconnected Graphene and H-BN Macroforms for TIM Applications

Vertically Aligned 1D and 2D Materials. Thermal property research of nanocarbon materials began with vertically aligned carbon nanotubes (VACNTs),^{166,167} inspired by their high thermal conductivity of over $3,000 \text{ W m}^{-1} \text{ K}^{-1}$ (Figure 9A).³⁹ However, thermal resistance between CNT array and Si interface is generally high, because of phonon mismatch between CNTs and Si substrate and the limited number of CNTs contacted with a substrate.^{168,169} It has been simulated that a 3D structure made up of CNTs and graphene with covalent bonding would significantly improve cross-plane thermal conductance.^{170,171} The concept of pillared structure was proposed in 2008, in which parallel CNTs are used to support graphene sheets as pillars, forming a perpendicular hybrid 3D carbon structure.¹⁷² Although synthesis is a major challenge, a seamless 3D carbon network was produced by CVD, showing covalent bonding between graphene and CNT knots (Figure 9B).^{173,174} However, very little research related to their thermal conductance

behavior was published, which is partially attributed to the difficulty in thermal conductivity measurements.

Densification of 2D Monoliths. Compression studies show that both compressed 3D graphene and 3D h-BN result in effective enhancement of their cross-plane thermal conductivity. The compression minimizes the pore size but maintains the wall thickness of graphene and h-BN, improving cross-plane thermal conductivity to $62\text{--}86\text{ W m}^{-1}\text{ K}^{-1}$ while keeping its surface conformity (Figure 9C).¹⁷⁵ 3D graphene aerogel by wet chemistry methods also draws great attention. Nevertheless, the pores are usually on the scale of hundreds of micrometers. A compressing procedure is therefore needed to squeeze out the air inside the macroform before the thermal conductivity can be improved (Figure 9D).^{162,176}

Graphene and h-BN Thin Films for Heat Spreaders. A heat spreader is used for dissipating heat from a hotspot to its surrounding area. In IC package, it is usually a thin copper or artificial graphite foil with high in-plane thermal conductivity. It has been demonstrated that a mechanically exfoliated few-layer graphene lowers the temperature of a GaN transistor by $\sim 20^\circ\text{C}$, showing great potential as a heat spreader as well as compatibility with current chip processing.¹⁷⁷ Freestanding reduced graphene oxide (rGO) paper can be fabricated by direct evaporation,¹⁷⁸ vacuum filtration,¹⁷⁹ and electrospray deposition¹⁸⁰ of GO dispersion, as well as surfactant-stabilized graphene nanoplates (GNP) solution.¹⁸¹ They usually show high thermal conductivity over $1,000\text{ W m}^{-1}\text{ K}^{-1}$, outperforming Cu and Al foils in terms of heat spreading capability. Thermal conductivity of the graphene film is effectively dependent on the sheet size of GO^{182,183} and its reduction process.^{184,185} Kumar et al. fabricated an rGO film by GO sheets with lateral size over $30\text{ }\mu\text{m}$, showing improved tensile strength (77 MPa) and thermal conductivity ($1,390\text{ W m}^{-1}\text{ K}^{-1}$) compared with smaller sizes.¹⁸⁶ In a recent work, Peng et al. created microfolds by mechanical pressing of microgasbags between graphene sheets, and these folds provide a graphene paper with mechanical flexibility even after high-temperature annealing ($3,000^\circ\text{C}$). The final graphene paper shows an ultrahigh thermal conductivity of $1,940\text{ W m}^{-1}\text{ K}^{-1}$,⁷⁴ exceeding the best performance of commercially used graphitized polyimide (PI) films.¹⁸⁷

As graphene films show limited thermal conductivity in the cross-plane direction,^{188,189} there have been attempts to build a freestanding 3D connected network to reach the balance between in-plane and cross-plane thermal conductivity.¹⁹⁰ Efforts have been made to increase the thermal conductivity at the cross-plane direction. *In situ* growth of carbon nanoring (CNR) between graphene sheets was reported to form a 3D bridged hybrid paper, and the cross-plane thermal conductivity showed a triple enhancement, reaching $5.8\text{ W m}^{-1}\text{ K}^{-1}$.¹⁹¹ Typical thermal conductivities of graphene-based heat spreaders are summarized in Table 2.

Graphene Fibers for Thermal Management. Graphene fibers can be prepared by wet spinning of concentrated GO dispersion followed by reduction. The outstanding mechanical and electrical properties have made graphene fibers good candidates for versatile applications,^{192,193} and the wet-spinning technique makes it easy for mass production up to kilometer scale.¹⁹⁴ For example, Xin et al. fabricated graphene fibers by wet spinning from mixed GO dispersion of large ($23\text{ }\mu\text{m}$) and small ($0.8\text{ }\mu\text{m}$) sheet size. The compact micro-nano hybrid structure enables the fiber to have high tensile strength of 1,080 MPa, achieving a high thermal conductivity of $1,290\text{ W m}^{-1}\text{ K}^{-1}$ after high-temperature treatments.¹⁹⁵ This is inspiring because lower thermal conductivity is found when using only large-sheet-sized GO at the

Table 2. Comparison of Thermal Conductivities of Typical Graphene Films

Materials	K ($\text{W m}^{-1} \text{K}^{-1}$)	Methods	Reference
rGO	1,100	direct evaporation, 2000°C annealed	Shen et al. ¹⁷⁸
rGO	1,390	vacuum filtration, large GO, hydriodic acid (HI) reduced	Kumar et al. ¹⁸⁶
rGO	1,434	electrospray deposition, 2,850°C annealed	Xin et al. ¹⁸⁰
rGO	1,940	scraping deposition, compressed and 3,000°C annealed	Peng et al. ⁷⁴
GNP	1,529	ball milling, 2,850°C annealed	Teng et al. ¹⁸¹
FLG	1,231 ($\kappa_{\parallel}$) 1.81 ($\kappa_{\perp}$)	Vacuum filtration, functionalized FLG	Liang et al. ¹⁷⁹
rGO + CNR	890 ($\kappa_{\parallel}$) 5.8 ($\kappa_{\perp}$)	Vacuum filtration, <i>in situ</i> growth of CNR (800°C)	Zhang et al. ¹⁹¹
EG + VACNT	255 ($\kappa_{\parallel}$) 35 ($\kappa_{\perp}$)	<i>in situ</i> growth of VACNT (850°C)	Qin et al. ¹⁹⁰

EG, expanded graphite; FLG, few-layered graphene.

same concentration, indicating the effectiveness of building the micro-nano hierarchical structure using GO with both large and small sheet sizes at an optimized proportion. It is worth noting that high-temperature annealing can significantly increase thermal conductivity, electrical conductivity, and Young's modulus of graphene. Simultaneously, it has been found that the graphene domain size grows to 780 nm at 2,850°C, which is one order of magnitude higher than that of PAN-based fibers (~ 20 nm at 2,700°C).¹⁹⁶ The greatly reduced phonon scattering at boundaries facilitates phonon transport and therefore effectively increases thermal conductivity. Further investigations are needed to make graphene fibers fully applicable in thermal management, acting as both fillers and thermal conductive wrappers in flexible devices.

Summary and Outlook

Thermal management has become an emerging topic with the rapid development of electronic and optoelectronic devices. In this review, we have presented an overview of 2D materials with their thermal physics, thermal measurements, and applications in thermal management. Although 2D materials with extraordinary electronic, optical, mechanical, and thermal properties promise numerous opportunities for future electronic, optoelectronic, and other devices, great challenges still exist when using 2D materials for thermal management applications. Here, we outline the three most prominent fields that may lead to breakthroughs in future thermal study and thermal applications of 2D materials.

Understanding Principles of Thermal Physics

2D materials give us a chance to study some fascinating thermal phenomena usually rarely detectable in bulk materials.^{197,198} As an example, a recent study discussed the significance of ballistic phonon transport and hydrodynamic phonon transport in graphene,³⁰ while other studies show the magnitude of ballistic transport in electronic and thermoelectric applications.¹⁹⁹ Moreover, it was calculated in a recent paper that the electronic contribution of thermal conductivity of graphene could be up to $\sim 10\%$, rather than a negligible value that was widely accepted previously.²⁰⁰ However, definitive experimental data supporting such a perspective is still lacking. In addition, it is widely accepted that an enhanced interfacial coupling would increase interface thermal conductance,⁶³ while precise measurements of interface thermal conductance are still a challenge, especially for atomically thick

van der Waals heterostructures.²⁰¹ A recent study proved that the solubility parameter may be the dominant factor for interface thermal conductance, the in-depth study of which may give clues on designing thermal interface and open a new avenue of understanding and controlling of heat flux in 2D materials and next-generation electronic devices.²⁰²

Controlled Growth and Assembly of 2D Materials into Designed Macrostructures

2D materials can be viewed as building blocks for assembly into 2D structures. Depending on the ways of assembly, different 3D macrostructures (i.e., thin films, foams, or 3D monoliths with defined structures) are useful in various thermal applications depending on heat dissipation requirements. In achieving maximum enhancement of thermal properties using 2D materials as TIM fillers, the construction of phonon conduits and growth of large domains of 2D materials are of great importance, as thermal conductivity is largely dependent on thermal percolation threshold and crystal domain sizes. In addition, 2D materials can be directly used as heat spreaders, for which controlled growth of 2D materials with preferable assembly on electronic devices is required. Vertically aligned graphene and CNT have shed light on the problem of c-axis thermal conductance; however, the thermal contact resistance between individual graphene flake and CNT is always a concern, which limits the performance in thermal management applications.^{203,204} Meanwhile, low-temperature CVD growth of graphene and h-BN on polymer substrate is always a popular topic for flexible electronic devices.²⁰⁵ Innovations in growing customized structures of 2D materials and their 3D macroforms within the specific context of application are much needed to develop thermal management applications for next-generation electronic devices.

Improving Thermal Measurement Methods

Despite great progress on measuring thermal properties of 2D nanomaterials, there are still questions that remain to be answered. Thermal conductivity data of samples with identical thickness may differ significantly from each other, varying from measurement to measurement and from sample to sample, which is not surprising because lateral sizes and environment of samples may also play an important role while it is difficult to precisely measure and control them.⁹⁰ It is still hoped that a novel thermal measurement method with lower errors of data and better adaptability to samples could be developed, with which researchers would be able to have a better understanding of fundamental thermal physics and a stronger tool to design thermal management applications.

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AUTHOR CONTRIBUTIONS

H.-M.C. and F.K. proposed the topic of this review. H.S. and J.L. investigated the literature and wrote the manuscript. J.W. and B.L. revised the manuscript.

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